Chronic tinnitus as phantom auditory pain

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OBJECTIVES: **To investigate similarities between patients who experience chronic tinnitus or pain and to formulate treatment strategies that are likely to be effective for patients who experience phantom auditory pain.**
STUDY DESIGN: **A total of 160 patients rated the severity and loudness of their tinnitus and completed the State-Trait Anxiety Inventory (STAI) and an abbreviated version of the Beck Depression Inventory (aBDI). Patients received counseling, audiometric testing, and matched the loudness of their tinnitus to sounds played through headphones.**
SETTING: **A specialized tinnitus clinic within an urban medical center.**
RESULTS: **Tinnitus severity was highly correlated with patients' degree of sleep disturbance, STAI, and aBDI scores. The self-rated (on a 1-to-10 scale)—but not the matched—loudness of tinnitus was correlated with tinnitus severity, sleep disturbance, STAI, and aBDI scores.**
CONCLUSIONS: **The severity of chronic tinnitus is correlated with the severity of insomnia, anxiety, and depression. These relationships are the same for many patients with chronic pain. Treatment recommendations are discussed in reference to these results.** (Otolaryngol Head Neck Surg 2001; 124:394-400.)

Tinnitus is the sensation of sound without external stimulation. Jastreboff[1] referred to tinnitus as phantom auditory perception. Outside of the auditory system, the most infamous example of phantom perception is reported by some patients who have lost a finger, hand, arm, toe, foot, or leg. These patients continue to perceive the presence of—and sometimes pain from—appendages that have been amputated. Missing appendages that continue to generate sensations are known as phantom limbs; painful sensations attributed to them are referred to as phantom limb pains.

Similarities between the perception of chronic tinnitus and the perception of chronic pain were listed by Tonndorf[2]:

- Both tinnitus and pain are subjective sensations;
- Both are continuous events that may change in quality and character over time;
- Both have the potential to be masked or reduced by appropriate sensory stimulation or medications;
- Both the auditory and somatosensory systems possess a well-developed network of efferent fibers that appear to exercise some control over afferent activity;
- Deafferentation (that is, a disruption in the balance between afferent and efferent activity) might explain both perceptions;
- Both perceptions are under the control of the central nervous system;
- Efforts to treat both sensations peripherally have met with limited success.

To this list of similarities Møller[3] added the following:

- Chronic pain and some forms of tinnitus are characterized by hypersensitivity to sensory stimulation;
- The anatomic locations of the neural structure(s) generating the sensations of chronic pain or tinnitus are different from the locations of the structures to which these symptoms are referred (the ears for tinnitus or the peripheral location of injury for pain);
- The strong psychological component that often accompanies chronic pain or tinnitus supports the hypothesis that brain areas (limbic/sympathetic) other than those responsible for sensory perception are involved;
- Pain and tinnitus are both heterogenous, multimodal disorders that can have different causes and pathophysiologies;
- Consequently, multimodal approaches should be used to treat these disorders.

Muhlnickel et al[4] used magnetoencephalography to compare the organization of auditory cortex in 10 chronic tinnitus patients with that of 15 nontinnitus control subjects. Results of their study demonstrated that the organization of auditory cortex in tinnitus patients was significantly different from the control subjects,

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especially in brain areas corresponding to perceived tinnitus frequencies. Muhlnickel et al[4] concluded that "similarities between these data and the previous demonstrations that phantom limb pain is highly correlated with cortical reorganization suggest that tinnitus may be an auditory phantom phenomenon."

Jeanmonod et al[5] hypothesized that positive neurological symptoms (including neurogenic pain and tinnitus) might be attributable to abnormal neuronal activity in the thalamus (specifically, low threshold calcium spike bursts that are related to thalamic cell hyperpolarization). A subsequent magnetoencephalographic study by Llinas et al[6] demonstrated that neurogenic pain and tinnitus are both characterized by thalamocortical dysrhythmia resulting from "inhibitory asymmetry between high- and low-frequency thalamocortical modules at the cortical level." These findings support the assertions of Jastreboff,[1] Tonndorf,[2] Møller,[3] and others who contend that abnormal asymmetries of neuronal activity are responsible for tinnitus generation.

It is clear that the perception of chronic tinnitus has many physiological characteristics in common with the perception of chronic pain. The studies mentioned here concentrated on similarities in neural mechanisms that result in the generation of tinnitus or pain. In his behavioral nosology, Briner[7] used the phrase "phantom auditory pain" to describe severe chronic tinnitus. The present study will evaluate psychological characteristics, reactions, and coincidental disturbances exhibited by patients who experience chronic tinnitus and compare these reactions or cosymptoms with those observed in pain patients. The goal is to contribute to the development of treatment strategies that are likely to be effective for patients experiencing phantom auditory pain.

METHODS

Detailed questionnaires were mailed to patients before their initial appointment at the Oregon Health Sciences University Tinnitus Clinic. These questionnaires requested information about patients' medical, hearing, and tinnitus histories. Appendix 1 contains 12 questions that constitute the Tinnitus Severity Index[8] (TSI), which is an efficient indicator of the negative impacts of tinnitus on patients. The State-Trait Anxiety Inventory (STAI)[9] and an abbreviated version of the Beck Depression Inventory (aBDI)[10] were also included.

The initial appointment at the clinic had the following format:

1. Patients met with staff members for an in-depth interview and review of their medical, hearing, tinnitus, and psychosocial histories and conditions. Patients received information and education about possible causes of their tinnitus as well as reassurance and counseling regarding factors that could exacerbate or improve their condition.
2. Audiological evaluations that included pure tone air and bone conduction thresholds; speech perception in quiet and noise; most comfortable loudness/uncomfortable loudness levels; and tympanometry.
3. Tinnitus evaluations that included matching tinnitus to sounds played through headphones; determination of minimum masking levels; and measurements of residual inhibition.
4. Various devices were demonstrated; evaluations of acoustic therapies were based on the patient's preferences and audiometric data. These could include hearing aids, in-the-ear sound generators, tinnitus instruments (combination hearing aid and sound generator), tabletop sound generation machines, tapes, CDs, or pillows with speakers.
5. Review of the results of evaluations; presentation of treatment plan and other recommendations.
6. Follow-up appointments or communication as needed for at least 6 months.

These protocols were reviewed and approved by the Institutional Review Board at the Oregon Health Sciences University. Informed consent was obtained in writing from patients before their participation in this study.

Data relating to patient demographics, audiometric thresholds, matched and self-rated (according to the 1-to-10 scale in Appendix 1) tinnitus loudness, tinnitus severity, sleep difficulties, aBDI, and STAI scores were analyzed (mean values were calculated and compared using analyses of variance and 2-tailed *t* tests; Pearson correlations were also calculated; appropriate statistical corrections were made as necessary).

RESULTS

We analyzed data from the last 160 consecutive patients (112 males, 48 females; mean age, 50.9 ± 12.8 years; age range, 17 to 87 years) who visited our clinic. Table 1 contains the grand averaged pure tone air conduction thresholds for these patients. This pattern of high-frequency sensorineural hearing loss is typical for our patient population.

Table 2 contains mean STAI, aBDI, tinnitus severity scores, and matched and self-rated tinnitus loudness values for 3 groups of patients based on their response to question 12: "Does your tinnitus interfere with sleep?" Note that mean values for all of these measures tend to increase with greater sleep interference. Statistically significant differences existed between the "No" and "Often" sleep interference groups on all measures except the matched loudness of tinnitus. Statistically significant differences existed between the "Sometimes" and

Table 1. Grand averages of pure tone air conduction thresholds of patients (dB HL)

	(Hz) 250	500	1000	2000	3000	4000	6000	8000
Right ear	18.3 ± 20.6	16.9 ± 21.1	19.7 ± 21.9	22.9 ± 24.1	33.1 ± 25.8	39.7 ± 27.2	43.6 ± 27.2	46.2 ± 28.5
Left ear	16.6 ± 15.2	14.7 ± 14.9	17.7 ± 17.3	22.2 ± 19.7	34.9 ± 22.8	40.3 ± 25.3	44.6 ± 25.4	45.9 ± 27.7

Table 2. Mean responses of patients grouped by frequency of sleep interference

	No Group 1 (n = 39)	Sometimes Group 2 (n = 73)	Often Group 3 (n = 48)
Tinnitus severity score	32.1 ± 4.7	40.1 ± 6.0	45.9 ± 7.6
Comparisons between groups	1 and 2	1 and 3	2 and 3
	$P \leq 0.0005$	0.0005	0.0005
Matched loudness of tinnitus (dB SL)	4.0 ± 2.8	7.5 ± 8.9	13.2 ± 12.4
Comparisons between groups	1 and 2	1 and 3	2 and 3
	$P \leq 0.619$	0.068	0.185
Self-rated loudness of tinnitus	5.8 ± 1.3	7.4 ± 1.6	8.0 ± 1.4
Comparisons between groups	1 and 2	1 and 3	2 and 3
	$P \leq 0.0005$	0.0005	0.146
State anxiety score	31.0 ± 11.7	37.3 ± 13.1	48.2 ± 15.4
Comparisons between groups	1 and 2	1 and 3	2 and 3
	$P \leq 0.144$	0.0005	0.002
Trait anxiety score	33.3 ± 9.6	38.4 ± 10.9	47.8 ± 12.6
Comparisons between groups	1 and 2	1 and 3	2 and 3
	$P \leq 0.166$	0.0005	0.002
Beck depression score	2.5 ± 2.1	5.5 ± 5.5	9.6 ± 6.7
Comparisons between groups	1 and 2	1 and 3	2 and 3
	$P \leq 0.070$	0.0005	0.004

"Often" sleep interference groups on all measures except the matched and self-rated loudness of tinnitus. Statistically significant differences existed between the "No" and "Sometimes" sleep interference groups on 2 measures: severity and self-rated loudness of tinnitus.

Table 3 contains mean STAI, aBDI, tinnitus severity scores, and matched and self-rated tinnitus loudness values for all of the patients in the study. Because there were no significant differences between male and female patients in any of these measures, Pearson correlation analyses were performed on values derived from the group as a whole. Fifty (31%; 30 males, 20 females) patients reported that they had current depression. Fifty-nine (37%; 35 males, 24 females) patients reported a history of depression. Scores on the aBDI ranged from 0 to 28 (maximum possible score, 39). Thirty-four (21%) patients scored 8 or higher on the aBDI. According to Dobie and Sullivan,[10] this can indicate that a patient is experiencing major depression.

Table 4 contains Pearson correlation coefficients and 2-tailed *P* values that resulted from statistical analyses of these measures. Note that tinnitus severity is highly correlated with STAI and aBDI scores. The self-rated, but not the matched, loudness of tinnitus is correlated with tinnitus severity, STAI, and aBDI. Both anxiety indexes were highly correlated with each other and also with the aBDI.

DISCUSSION

Results from this and other studies demonstrated that the severity of chronic tinnitus is often correlated with insomnia,[11] anxiety,[12] and depression.[13] As illustrated in Fig 1, these symptoms can form a vicious circle and exacerbate each other. Insomnia, anxiety, and depression are also common co-symptoms for patients with chronic pain. In fact, the word "pain" can be substituted for the word "tinnitus" in Fig 1 and the relationships among these symptoms will remain the same.

What other characteristics do pain patients have in common with tinnitus patients? Numerous studies contributed to the following list: hypochondriasis; obsessive-compulsive tendencies; high degrees of self-focus and self-attention; perceived lack of control over symptoms and life events; catastrophic thinking; focusing or dwelling on symptoms; maladaptive coping strategies; reluctance to admit to problems other than immediate

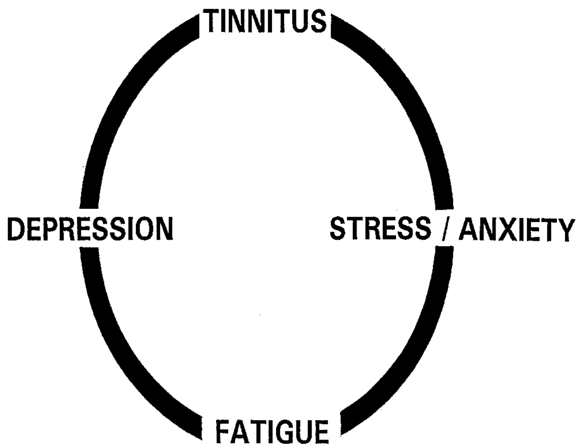

Fig 1. A vicious circle of symptoms.

Table 3. Means and standard deviations of measures

Measures	Males (n = 112)	Females (n = 48)
Tinnitus severity score	37.60 ± 9.30	38.86 ± 9.50
Matched loudness of tinnitus (dB SL)	5.91 ± 6.87	11.97 ± 10.37
Self-rated loudness of tinnitus	6.94 ± 1.67	7.08 ± 1.64
State anxiety score	36.73 ± 14.84	39.71 ± 15.11
Trait anxiety score	36.90 ± 12.46	41.19 ± 12.33
Beck depression inventory	4.98 ± 5.40	5.73 ± 5.79

physical symptoms; the perceived severity of the patient's condition is not necessarily related to objective measures of stimulus intensity; and the severity of symptoms can be related to patients' perceptions of attitudes or reactions of others to their condition. Of course, every patient does not necessarily possess any or all of these characteristics. However, these traits are more likely to occur in pain or tinnitus patients who perceive their symptoms to be severe or debilitating.

Did the onset of chronic tinnitus cause these behaviors or co-symptoms to occur? Dobie and Sullivan[10] reported that approximately 50% of their tinnitus patients with depression had at least 1 bout of major depression before the onset of their tinnitus. Rizzardo et al[14] reported that 50% of their patients exhibited psychological symptoms before the onset of tinnitus; 71% of these patients experienced greater than normal levels of depression, anxiety, hypochondriasis, and/or neuroticism after tinnitus began.

Rizzardo et al[14] stated that there appears to be a "link between psychological distress and tinnitus in a potential somatopsychological and psychosomatic vicious circle (a psychological predisposition to react emotionally to events, tinnitus as a source of distress that reinforces the symptom, accentuating hypochondriac fears)." Dobie and Sullivan[10] agree that some people are more predisposed to depression than others and that tinnitus is one of many internal and external triggers that can precipitate major depression in susceptible individuals. Perhaps the most logical conclusion was stated by Halford and Anderson[12]: "It is considered that the causal relationship between these psychological variables and tinnitus severity is likely to be bi-directional."

How can this information be used to help patients with severe chronic tinnitus? Because tinnitus patients share many similarities with chronic pain patients, otolaryngology clinicians can use some of the same techniques and strategies in tinnitus treatment that are used in pain management. These include the following[15]:

1. Treatment of depression with medications and/or psychotherapy. Sullivan et al[16] demonstrated that successful treatment of depression can reduce the severity of tinnitus for patients experiencing both maladies. Some antidepressant medications will also improve sleep patterns and reduce anxiety. Identification of tinnitus patients who are also experiencing depression can be accomplished by using the complete Beck Depression Inventory[17] or other appropriate instruments (such as the aBDI[10]).
2. Treatment of insomnia with medications, relaxation therapy, and/or acoustic therapy—this includes pleasant sounds generated in the bedroom by tabletop devices (such as Sound Spa model SS-400B made by Homedics Inc, Commerce Township, MI), tapes or CDs (such as the Tinnitus Relief System distributed by Personal Growth Technologies, Pleasant Ridge, MI), pillows with speakers (like the Sound Pillow distributed by Phoenix Productions, San Antonio, TX), fans, or small fountains.
3. Treatment of anxiety with medications, relaxation therapy, psychotherapy, biofeedback, hypnosis, massage, or any other appropriate stress management techniques.
4. Any neuroses, psychoses, or other maladaptive behaviors need to be assessed and addressed during a series of psychotherapy or counseling sessions. Many experts agree with House[18] who wrote that "most tinnitus patients can often be helped by psychological intervention." If the physician, nurse, or audiologist does not feel that they have the time or training to provide the counseling personally, the clinician should refer the patient to an appropriate mental health professional.

Acoustic therapy is one way to give patients some control over and relief from their tinnitus. This can

Table 4. Pearson correlation coefficients

	Anxiety state	Anxiety trait	Beck depression inventory	Matched loudness	Self-rated loudness
Tinnitus severity	0.573***	0.567***	0.559***	0.085	0.523***
State anxiety		0.851***	0.800***	0.062	0.273**
Trait anxiety			0.837***	0.051	0.255*
Beck depression inventory				0.145	0.291**

*$P < 0.005$; **$P < 0.001$; ***$P < 0.0001$.

Table 5. Responses to questionnaires (n = 174 patients)

	Initial	Follow-up	$P \leq$
Self-rated loudness of tinnitus	6.69 ± 2.02	6.42 ± 2.18	0.117
Does your tinnitus...			
Make you feel irritable or nervous?	3.27 ± 0.97	2.88 ± 0.87	0.0005*
Make you feel tired or stressed?	3.25 ± 1.13	2.91 ± 0.95	0.0005*
Make it difficult for you to relax?	3.37 ± 1.03	3.06 ± 1.01	0.0005*
Make it uncomfortable to be in a quiet room?	3.43 ± 1.24	3.31 ± 1.26	0.164
Make it difficult to concentrate?	3.30 ± 1.05	3.03 ± 0.97	0.001*
Make it harder to interact pleasantly?	2.97 ± 1.11	2.74 ± 0.96	0.014
Interfere with required activities?	2.86 ± 1.21	2.59 ± 1.03	0.004*
Interfere with social activities?	3.01 ± 1.20	2.82 ± 1.06	0.062
Interfere with your overall enjoyment of life?	3.37 ± 1.10	3.12 ± 1.10	0.004*
Interfere with sleep?	2.11 ± 0.77	1.90 ± 0.77	0.0005*
How much effort is it to ignore tinnitus?	2.83 ± 0.98	2.47 ± 0.91	0.0005*
How much discomfort do you usually experience when tinnitus is present?	2.86 ± 1.00	2.50 ± 0.92	0.0005*

*Statistically significant improvement.

include the aforementioned devices as well as in-the-ear sound generators (such as those made by Starkey, Eden Prairie, MN, or by General Hearing Instruments, Harahan, LA), hearing aids, or combination instruments (hearing aid plus sound generator).

Because patients with severe tinnitus often have negative affectivity (characterized by tendencies to be distressed, worried, anxious, and self-critical), their counseling should be as positive and productive as possible. Jakes et al[19] admonished clinicians: “instead of advising patients that they must ‘learn to live with it’ with no advice as to how this is to be achieved, one could rather advise them that distress about tinnitus is not determined by having tinnitus, and that an intrusive, subjectively loud tinnitus will not necessarily produce a strong effect on the patient’s social, domestic, or economic functioning.” After appropriate tests have ruled out acoustic neuroma or other retrocochlear causes for a patient’s tinnitus, clinicians should reassure the patient that tinnitus is usually related to hearing loss[20] and that it is a harmless perception of sound generated by the auditory system. Tinnitus will not necessarily become worse with time, and it does not portend additional hearing loss or the manifestation or exacerbation of any other medical condition.

Because each tinnitus patient has a unique medical, psychological, and social history, therapeutic interventions should be individualized. In fact, the most successful treatment programs use multimodal strategies that are designed to address the specific needs of each patient. Hawthorne et al[21] concluded that psychiatric intervention “significantly reduced the emotional distress” in a population of tinnitus patients. “This was achieved by not only dealing with the somatic disease but also by psychiatric management of the coincidental mental distress. This was very time-consuming. Many of the patients had complex difficulties; although they all had tinnitus and most had mood disturbance, no history was typical. The problems were protean and the psychotherapeutic interventions had to be tailored for each person.”

How effective are individualized, multimodal treatment programs at reducing the severity of chronic tinnitus? We conducted a long-term follow-up study of 174 patients (130 males, 44 females; mean age, 55.9 years) who were evaluated and treated in our clinic between 1994 and 1997.[22] One to 4 years after their initial clinic appointment (mean, 2.3 years), these patients reported no significant change in self-rated loudness of tinnitus.

However, there was statistically significant improvement in 9 of the 12 measures of tinnitus severity (feeling irritable or nervous; feeling tired or stressed; difficulty relaxing; difficulty concentrating; interference with required activities; interference with overall enjoyment of life; interference with sleep; the amount of effort to ignore tinnitus; and the amount of discomfort usually experienced when tinnitus is present) for the entire patient population (Table 5). A subset of 40 patients who purchased and used in-the-ear devices (hearing aids, sound generators, or combination instruments) reported significant improvement in all 12 measures of tinnitus severity.

If a clinician has assessed and treated every reasonable medical cause for a patient's tinnitus, and the patient reports little improvement in tinnitus severity, the clinician should do 1 of 2 things: (1) spend the time necessary to effectively treat the patient according to procedures described here and elsewhere[23]; or (2) refer the patient to a comprehensive treatment center with experienced personnel who are willing and able to spend a substantial amount of time with each patient. For a certain number of patients with phantom auditory pain, only a specialized treatment program of this type can help them to improve their condition. Telling patients that because "nothing can be done" for tinnitus they just have to "learn to live with it" is both erroneous and counterproductive.

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Appendix 1.
Tinnitus severity questions

DIRECTIONS: For the questions below, please CIRCLE the number that best describes you

	Never	Rarely	Sometimes	Usually	Always
Does your tinnitus					
1. Make you feel irritable or nervous	1	2	3	4	5
2. Make you feel tired or stressed	1	2	3	4	5
3. Make it difficult for you to relax	1	2	3	4	5
4. Make it uncomfortable to be in a quiet room	1	2	3	4	5
5. Make it difficult to concentrate	1	2	3	4	5
6. Make it harder to interact pleasantly with others	1	2	3	4	5
7. Interfere with your required activities (Work, home, care, or other responsibilities)	1	2	3	4	5
8. Interfere with your social activities or other things you do in your leisure time	1	2	3	4	5
9. Interfere with your overall enjoyment of life	1	2	3	4	5

10. Does your tinnitus interfere with sleep?
   - No................................ 1
   - Yes, sometimes............ 2
   - Yes, often..................... 3

11. How much of an effort is it for you to ignore tinnitus when it is present?
   - Can easily ignore it.............................. 1
   - Can ignore it with some effort............. 2
   - It takes considerable effort................... 3
   - Can never ignore it............................... 4

12. How much discomfort do you usually experience when your tinnitus is present?
   - No discomfort.............................. 1
   - Mild discomfort........................... 2
   - Moderate discomfort................... 3
   - A great deal of discomfort........... 4

On the scale below, please CIRCLE the number that best describes the loudness of your usual tinnitus

1	2	3	4	5	6	7	8	9	10
Very quiet				Intermediate					Very loud